Pre-Harvest Agricultural Forecasting: A Predictive Model for Palay Yields

Jose Rizal University · SDG 2: Zero Hunger — Target 2.4

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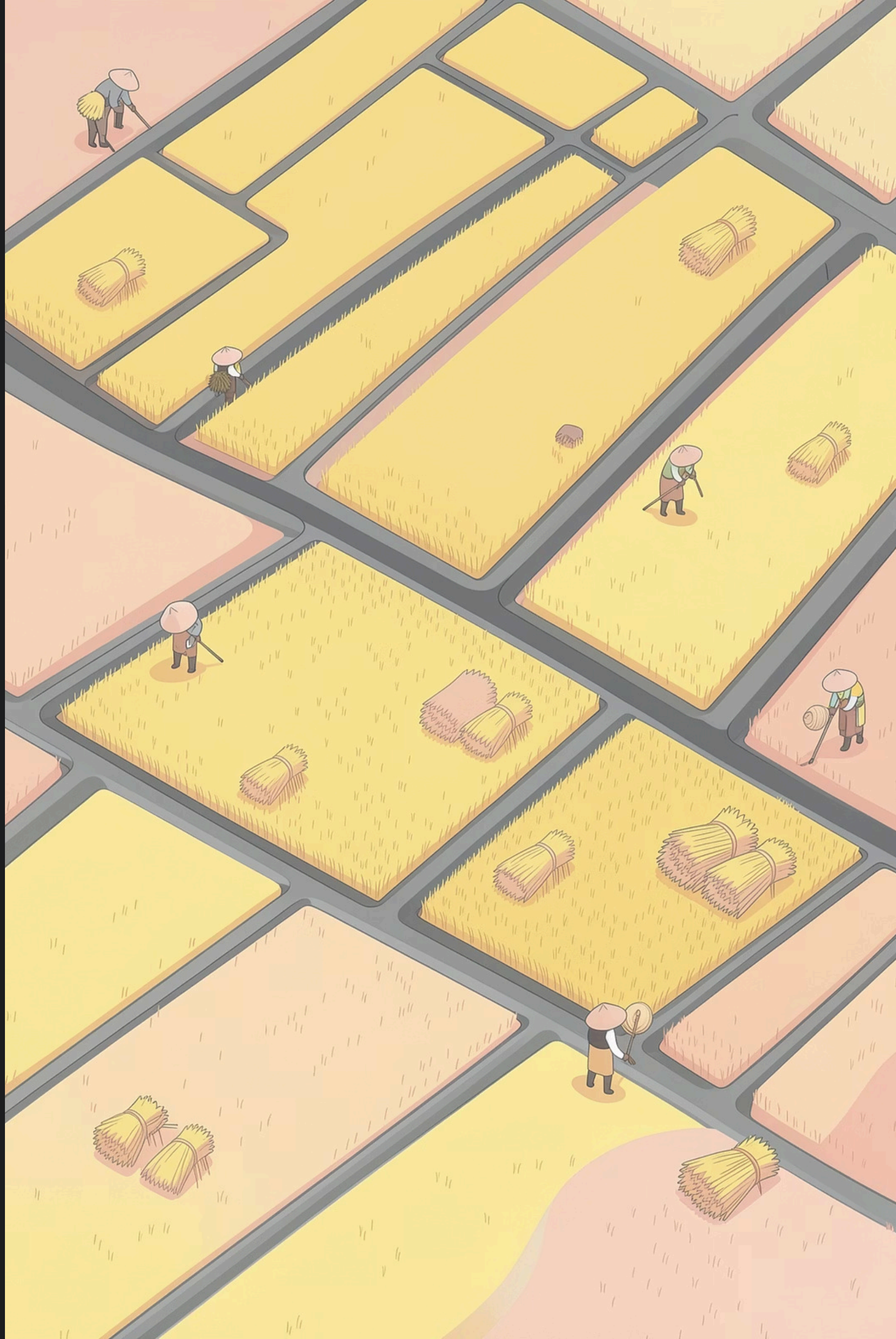
PSA DATA · MLR MODEL



The Problem: A Fragile Supply Chain

National Baseline (2023)

The Philippines produced **20.06 million metric tons** of palay in 2023 (PSA, 2024) — yet supply remains dangerously concentrated.

- Central Luzon (Region III): 3.64M MT — **18.1%** of national supply
- Cagayan Valley (Region II): 3.03M MT
- Together, just two regions control **33%** of the country's entire rice supply

The Compounding Threat

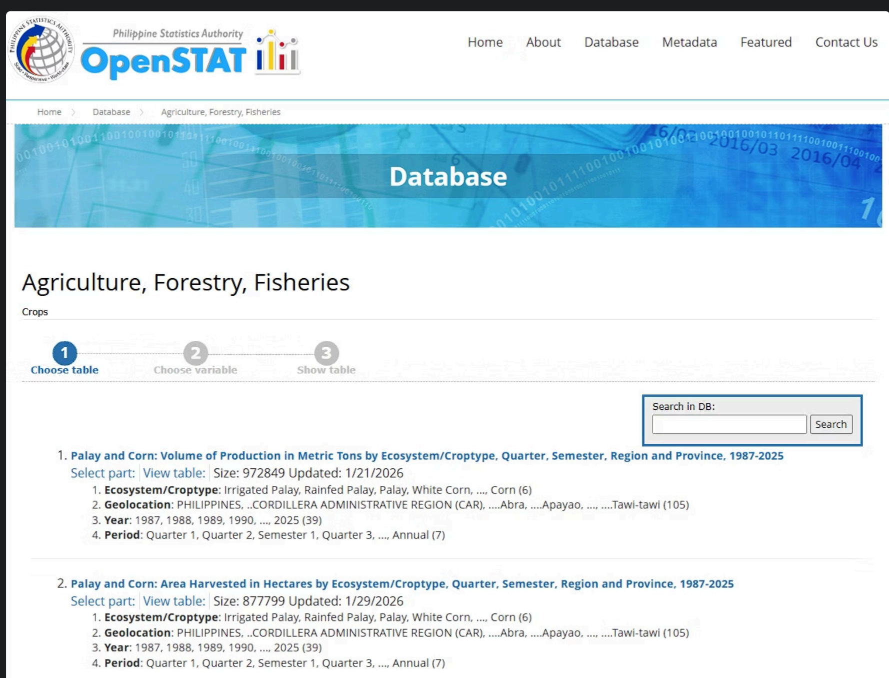
Climate volatility from El Niño, combined with soaring global fertilizer costs, caused farmgate prices to surge nearly **32% in early 2024** (DOST-PCAARRD, 2024).

⚠ The objective: forecast exact production volume to preemptively trigger government safety nets — before regional shortages cause national price spikes.

DATASET OVERVIEW

The Dataset Blueprint

Source: Philippine Statistics Authority (PSA) — OpenSTAT Portal · **Scope:** Quarterly, 2003–2024 · 2,400+ historical rows across all Philippine provinces



Variable Structure

X_1 — Area Harvested

Total land area harvested (Hectares)

X_2 — Fertilizer - Urea Applied

Urea volumes applied

X_3 — Fertilizer - Ammosul Applied

Ammosul volumes applied

X_4 — Ecosystem

Irrigated vs. Rainfed classification

X_5 — Fertilizer - Complete (14-14-14) Applied

Complete fertilizer volumes applied

X_6 — Harvest Quarter

Seasonality Proxy

Y — Target Variable

Volume of Production (Metric Tons)

Model Selection: Multiple Linear Regression

Why MLR?

Continuous Target

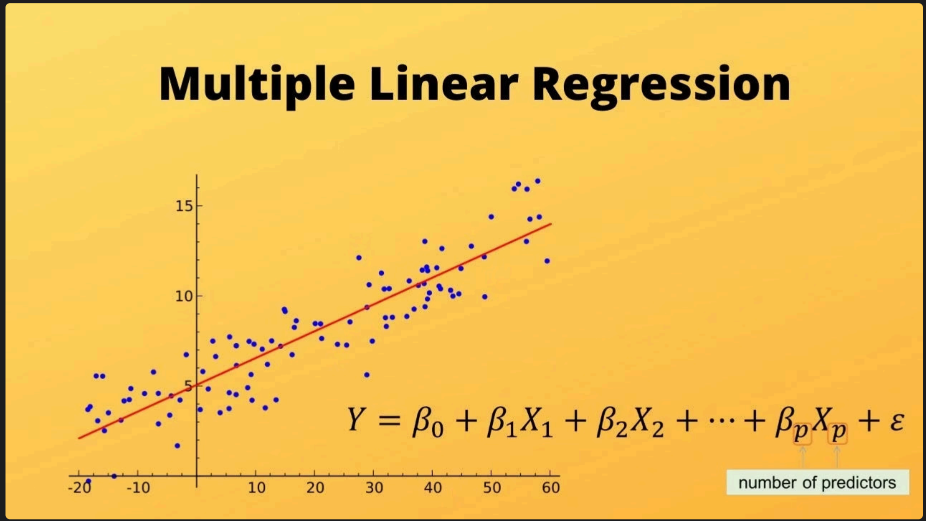
Production volume in metric tons is inherently continuous — MLR is the statistically appropriate choice.

Full Interpretability

Unlike "black box" models, MLR provides exact numerical weights for every feature — readable by any policymaker.

Literature Grounding

Lagrazon & Tan (2023) achieved $R^2 = 0.968$ using Gaussian Process Regression. MLR sacrifices marginal precision for total transparency.



Transparency was prioritized so government units can audit and trust every coefficient in the model.

From Forecast to Policy Action

$$Y(\text{Yield}) = \beta_0 + \beta_1(\text{Hectares}) + \beta_2(\text{Urea}) + \beta_3(\text{Ammosul}) + \beta_4(\text{Ecosystem}) + \beta_5(\text{Complete}) + \beta_6(\text{Quarter}) + \epsilon$$

1

Current Quarter Inputs

Planted hectares, fertilizer supply, and ecosystem type are recorded at the start of the season.

2

Model Projection

The equation instantly computes projected yield months before the physical harvest occurs.

3

Pre-emptive Action

Safety nets — subsidies, buffer stocks, import adjustments — are triggered *before* shortages materialize.

Fulfilling SDG 2.4: Mathematically auditing agricultural capability to build a resilient, climate-adaptive food supply chain across all Philippine regions.

References

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Lagrazon & Tan (2023)

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DOST-PCAARRD (2024)

2024 Rice production and price trends, decreasing imports, and initiatives to improve yield.

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